
HW2: Transfer Function Representation in MATLAB

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Use MATLAB to generate the following transfer function two different ways

$$G(s) = \frac{3}{s^3+9s^2+20s}$$

$$H(s) = \frac{10(s+2)}{s^2(s+4)(s+5)}$$

Ratio of factors

By defining the roots of the numerator (zeros) and denominator (poles)

```
Gzpk = zpk([],roots([1 9 20 0]),3)
Hzpk = zpk([-2],[0,0,-4,-5],10)
```

Gzpk =

$$\frac{3}{s(s+5)(s+4)}$$

Continuous-time zero/pole/gain model.

Hzpk =

$$\frac{10(s+2)}{s^2(s+4)(s+5)}$$

Continuous-time zero/pole/gain model.

Ratio of polynomials

By defining the numerator and denominator as polynomials Uses nested conv() functions

```
Gtf = tf(3,[1 9 20 0])
```

```
Htf = tf(10*[1 2],conv(conv([1 0 0],[1 4]),[1 5]))
```

Gtf =

$$\frac{3}{s^3 + 9 s^2 + 20 s}$$

Continuous-time transfer function.

Htf =

$$\frac{10 s + 20}{s^4 + 9 s^3 + 20 s^2}$$

Continuous-time transfer function.

Bonus - as function of 's'

Can also define a transfer function directly as a function of 's'

```
s = tf('s');  
Gs = 3/(s^3+9*s^2+20*s)  
Hs = 10*(s+2)/(s^2*(s+4)*(s+5))
```

Gs =

$$\frac{3}{s^3 + 9 s^2 + 20 s}$$

Continuous-time transfer function.

Hs =

$$\frac{10 s + 20}{s^4 + 9 s^3 + 20 s^2}$$

Continuous-time transfer function.

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